

Work Energy and Power

• Workdone

→ Work is said to be done if force applied on a body displaces the body in any direction except perpendicular to the applied force.

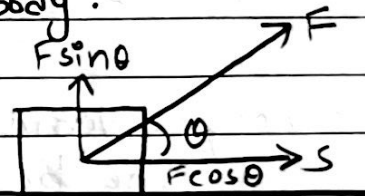
• Workdone by constant force

→ The work done by a constant force is given by the scalar product of applied force and displacement of the body.

i.e. $W = \vec{F} \cdot \vec{s}$

$\therefore \boxed{W = F s \cos \theta}$

GSS_Notes

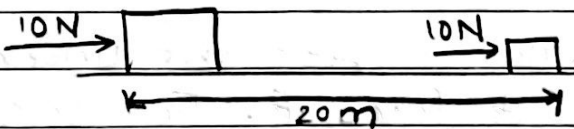


Case (i) When $\theta = 0^\circ$ then $W = Fs$ (maximum)

case (ii) When $\theta = 90^\circ$ $W = 0$

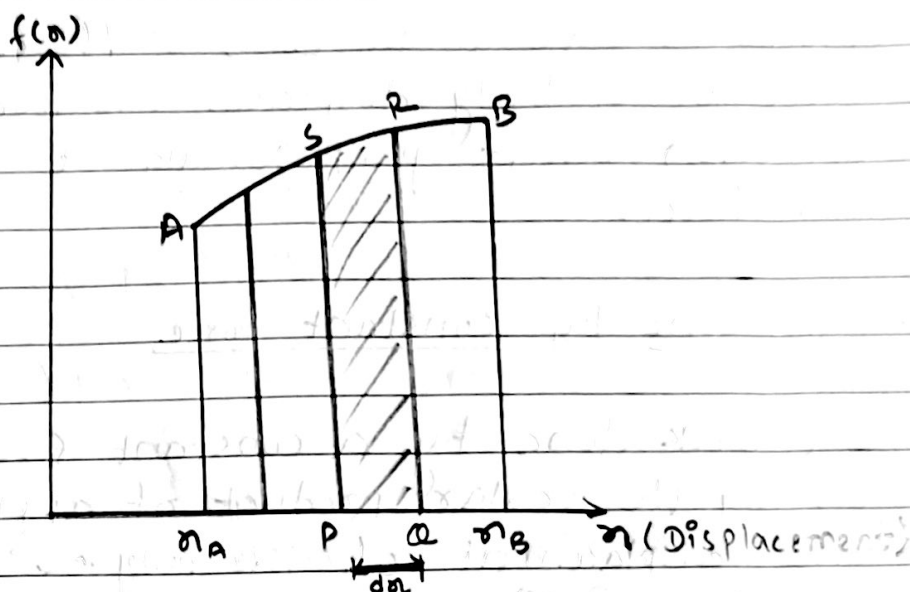
case (iii) When $\theta = 180^\circ$ then $W = -Fs$

• Work done by Variable Force



$W = Fs = 200 \text{ J}$

The force which changes with the change in position of body is called Variable Force.



Consider a variable force acting on a body displaces the body from x_a to x_b which is represented by the curve AB in the force-displacement graph.

Let us divide total displacement into large number of small displacement so that the force for a small displacement can be considered as constant.

Let $PQ = dx$ be a small displacement under the action of constant force F . Then small workdone is given by.

$$dW = F dx = \text{Area of Strip PQRS} \rightarrow \textcircled{1}$$

Then Total workdone can be obtained by

integrating such small workdone from $x=x_A$ to $x=x_B$.

$$\therefore W = \int_{x_A}^{x_B} f dx = \text{Area between the curve AB and displacement axis.}$$

- ① A block of weight 150 N is pulled 20 m along a horizontal surface at constant velocity. Calculate the workdone by pulling force if the coefficient of kinetic friction is 0.20 and the pulling force makes an angle of 30° with the horizontal.

→ Soln,

$$W = 150 \text{ N}$$

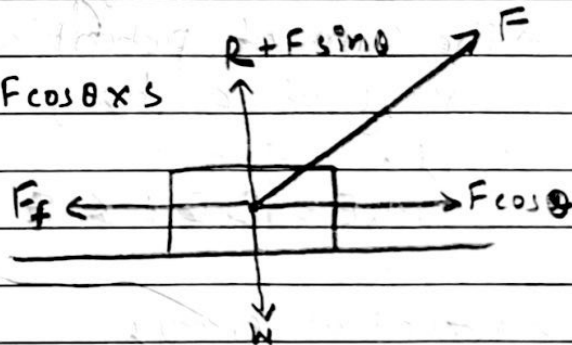
$$S = 20 \text{ m}$$

$$W = ?$$

$$\mu = 0.20$$

$$\theta = 30^\circ$$

$$W = F \cos \theta \times S$$



Now,

GSS_Notes

$$F \cos \theta = F_f$$

$$\text{or, } F \cos \theta = \mu R$$

$$\text{or, } F \cos \theta = \mu (W - F \sin \theta)$$

$$\text{or, } F \cos 30 = 0.20 (150 - F \sin 30)$$

$$\text{or, } \frac{\sqrt{3}}{2} F = 0.20 \left(150 - F \times \frac{1}{2} \right)$$

$$\text{or, } \frac{\sqrt{3}}{2} F = 30 - \frac{1}{10} F$$

$$\text{or, } 0.96 F = 30$$

$$\text{or, } F = 31.05 \text{ N}$$

$$\therefore F = 31.05 \text{ N}$$

Then,

$$\begin{aligned} W &= F \cos \theta \times s \\ &= 31.05 \times \frac{\sqrt{3}}{2} \times 20 \\ &= 537.89 \text{ J} \end{aligned}$$

- Energy

→ The capacity of doing work is called energy.

The S.I Unit of energy is Joule.

- > Mechanical Energy

→ The sum of potential energy and kinetic energy of a body is called mechanical energy.

$$\therefore M.E = P.E + K.E$$

- > Potential energy

→ The energy possessed by a body by virtue of its position is called potential energy.

The P.E of a body of mass m at a height h is given by

$$P.E = mgh$$

- > Kinetic energy

→ The energy possessed by a body by virtue of its motion is called kinetic energy.

The kinetic energy of a body of mass m moving with velocity v is given by

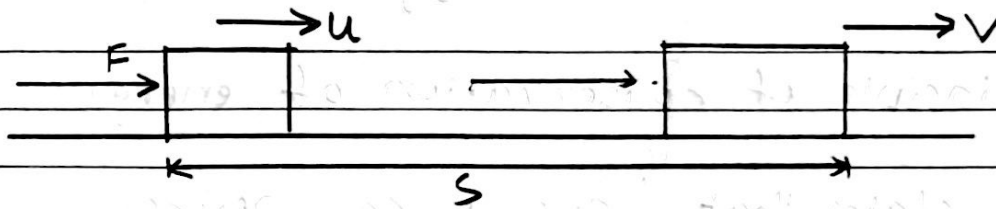
$$K.E = \frac{1}{2} mv^2$$

- Work-energy theorem

↳ It states that, "The workdone by a force acting on a body is equal to the change in kinetic energy of the body."
i.e. Workdone = change in K.E

$$\therefore W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

Proof



Consider a constant force F is acting on a body of mass m . Let s be the displacement of the body. Let u be the initial velocity and v be the final velocity of the body.

GSS_Notes

Now workdone by the force on the body is given by

$$W = Fs$$

$$\therefore W = mas \rightarrow (i)$$

Where a is the acceleration of the body
Now,

From the equation of motion, we have

$$v^2 = u^2 + 2as$$

$$\text{or, } v^2 - u^2 = 2as$$

$$\text{or, } as = \frac{v^2 - u^2}{2}$$

Putting this value in eqn (i), we get

$$W = m \left(\frac{v^2 - u^2}{2} \right)$$

$$W = \frac{mv^2}{2} - \frac{mu^2}{2}$$

$$\therefore W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

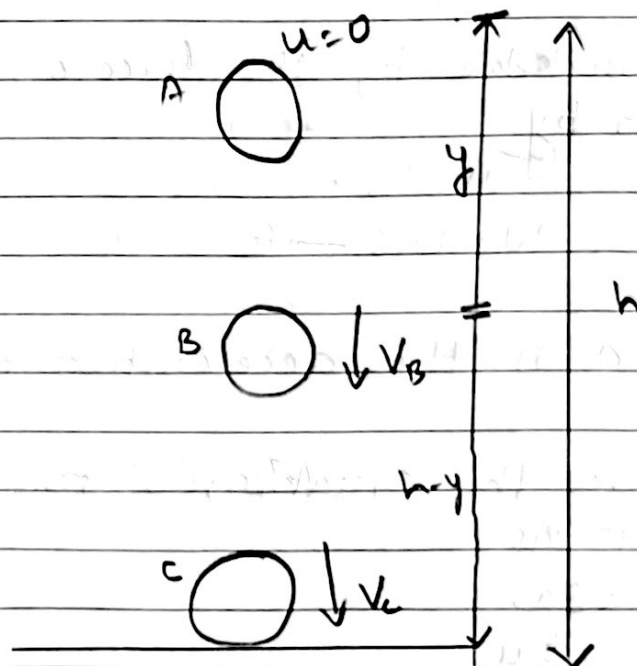
i.e. Work done = Change in K.E

This proves work-energy theorem.

• Principle of conservation of energy

→ It states that, "Energy can neither be created nor can be destroyed but it can be changed from one form to another."
i.e.: Total energy of an isolated system remains constant.

Proof.



Consider a body of mass m is falling freely from a ~~body~~ height h . i.e. from position A. Suppose the body reaches to the position B by travelling a distance y and finally reaches to the position C. i.e. ground by travelling further distance $h-y$ as shown in figure.

At position A.

$$(P.E)_A = mgh$$

$$\text{and } (K.E)_A = \frac{1}{2} mu^2 = 0 \quad [\because u=0]$$

Then total energy or mechanical energy is given by

$$\begin{aligned}(T.E)_A &= (P.E)_A + (K.E)_A \\ &= mgh + 0 \\ &= mgh \rightarrow \textcircled{i}\end{aligned}$$

GSS_Notes

At position B

$$\begin{aligned}(P.E)_B &= mg(h-y) \\ &= mgh - mgy\end{aligned}$$

Let v_B be the velocity of body at position B.

Now,

Applying the equation of motion

$v^2 = u^2 + 2as$ between the position A & B
we get

$$V_B^2 = 0 + 2gy$$

$$\therefore V_B^2 = 2gy$$

Then,

$$(K.E)_B = \frac{1}{2} m v_B^2$$

$$= \frac{1}{2} \times m \times 2gy$$

$$= mgy$$

Now,

$$\begin{aligned} \text{#a: (Total energy)}_B &= (P.E)_B + (K.E)_B \\ &= mgh - \cancel{mgy} + \cancel{mgy} \\ &= mgh \quad \text{--- (2)} \end{aligned}$$

At position c

$$(P.E)_c = mg \times 0 = 0 \quad [\because h=0]$$

✳

Let V_c be the velocity of body at position c.

Now, applying the equation of motion $V^2 = u^2 + 2as$ between the positions A & c, we get,

$$V_c^2 = 0 + 2gh$$

$$\therefore V_c^2 = 2gh$$

Then,

$$(K.E)_c = \frac{1}{2} m v_c^2$$

$$= \frac{1}{2} \times m (2gh)$$

$$= mgh$$

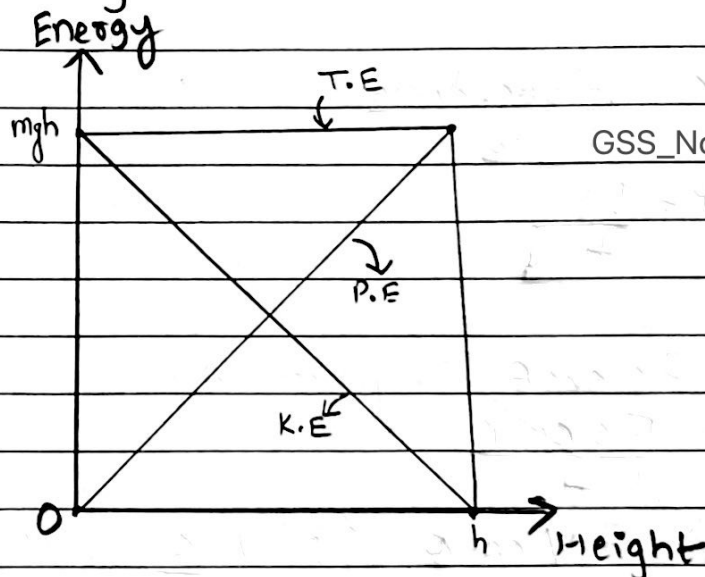
Now,

$$\begin{aligned} (\text{Total Energy})_c &= (P.E)_c + (K.E)_c \\ &= 0 + mgh \\ &= mgh \quad \text{--- (3)} \end{aligned}$$

Here, From eqns (1), (2) & (3) we have
 $(T.E)_A = (T.E)_B = (T.E)_c$

This shows that total ~~ener~~ energy of a freely falling body ~~is~~ remains same at all the positions, This proves principle of conservation of energy.

The variation of potential energy and kinetic energy of a freely falling body with height is as shown in fig below.



* Relation between Linear momentum and kinetic energy

→ Since, $K.E = \frac{1}{2} mv^2$

$$= \frac{1}{2} \frac{m^2 v^2}{m}$$

$$= \frac{1}{2} \frac{(mv)^2}{m}$$

$$\boxed{K.E = \frac{p^2}{2m}}$$

$$\Rightarrow K.E \propto \frac{1}{m}$$

$$\therefore \boxed{\frac{K.E_1}{K.E_2} = \frac{m_2}{m_1}}$$

* Power

→ The rate of doing work is called Power.

$$\therefore \text{Power} = \frac{\text{Work done}}{\text{time}}$$

$$\text{i.e.} \Rightarrow \boxed{P = \frac{W}{t}}$$

Since,

$$W = Fs \cos \theta$$

$$\text{So, } P = \frac{Fs \cos \theta}{t}$$

$$\therefore P = FV \cos \theta \quad \left\{ \because V = \frac{s}{t} \text{ is velocity} \right\}$$

When $\theta = 0^\circ$ then

$$\boxed{P = FV}$$

- ⑥ A stationary mass explodes into two parts of mass 4 kg and 40 kg. The initial kinetic energy of larger mass is 10 J. Find the velocity of smaller mass.

→ Soln,

$$m_1 = 4 \text{ kg}$$

$$m_2 = 40 \text{ kg}$$

$$K.E_2 = 10 \text{ J}$$

$$V_1 = ?$$

$$\text{Since, } \frac{K.E_1}{K.E_2} = \frac{m_2}{m_1}$$

$$\text{or, } \frac{K.E_1}{10} = \frac{40}{4}$$

$$\text{or, } K.E_1 = 100 \text{ J}$$

$$\text{or, } \frac{m_1 V_1^2}{2} = 100 \text{ J}$$

$$\text{or, } 4 \times V_1^2 = 200$$

$$\text{or, } V_1^2 = 50$$

$$\text{or, } V_1 = \sqrt{50}$$

$$\therefore V_1 = 7.07 \text{ m/s}$$

- ⑦ A train of mass $2 \times 10^5 \text{ kg}$ moves at constant speed of 72 km/hr up a straight inclined against a frictional force of $1.28 \times 10^4 \text{ N}$. The incline is such that the train rises vertically 1m for every 100m travelled along the incline. Calculate the necessary power developed by the train.

→ Soln,

$$m = 2 \times 10^5 \text{ kg}$$

$$V = 72 \text{ km/hr} = \frac{72 \times 1000}{60 \times 60} \text{ m/s} = 20 \text{ m/s}$$

$$F_f = 1.28 \times 10^4 \text{ N}$$

$$\sin \theta = \frac{1}{100}$$

$$P = ?$$

Now,

$$P = FV$$

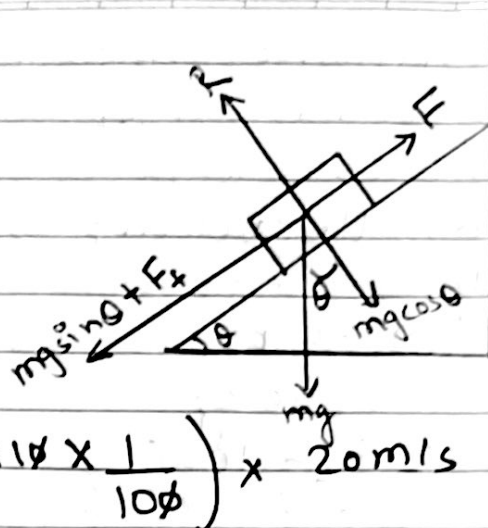
$$\text{or, } P = (F_f + mg \sin \theta) V$$

$$\text{or, } P = \left(1.28 \times 10^4 + 2 \times 10^5 \times 10 \times \frac{1}{100} \right) \times 20 \text{ m/s}$$

$$\text{or, } P = 32800 \times 20$$

$$\text{or, } P = 656000$$

$$\therefore P = 656000 \text{ W}$$



- ⑤ A water reservoir tank of capacity 250 m^3 is situated at a height of 20 m from the water level. What will be the power of an electric motor to be used to fill the tank in 3 hrs? Efficiency of motor is 70% .

→ Soln,

$$\text{Volume} = 250 \text{ m}^3 = V$$

$$\text{Height} = 20 \text{ m} = h$$

$$P_{\text{out}}$$

$$P_{\text{in}} = ?$$

$$t = 3 \text{ hrs} = 3 \times 60 \times 60 = 10800 \text{ s}$$

$$\eta = 70\%$$

Now,

$$P_{\text{out}} = \frac{W}{t}$$

$$= \frac{mgh}{t}$$

$$= \frac{\rho V g h}{t} \left[\text{density } (\rho) = \frac{m}{V} \right]$$

$$= \frac{1000 \times 250 \times 10 \times 20}{10800}$$

$$P = 4629.62 \text{ W}$$

Again,

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\%$$

$$\text{or, } 70\% = \frac{4629.62}{P_{\text{in}}} \times 100\%$$

$$\text{or, } P_{\text{in}} = \frac{4629.62}{7} \times 10$$

$$\text{or, } P_{\text{in}} = 6613.74 \text{ W}$$

$$\therefore P_{\text{in}} = 6613.74 \text{ W}$$

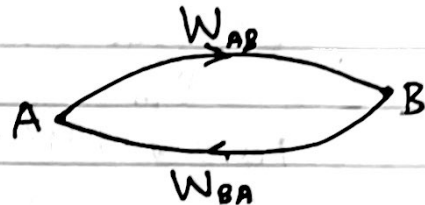
GSS_Notes

• Conservative and non-conservative forces

* Conservative forces

→ A force is said to be conservative if total work done by the force on the body over a closed path is zero.

$$\text{i.e. } \sum W = 0$$
$$\text{or, } W_{AB} + W_{BA} = 0$$



Work done by conservative force doesn't depend upon the length of path followed by the body but it only depends upon the initial and Final position of the body.

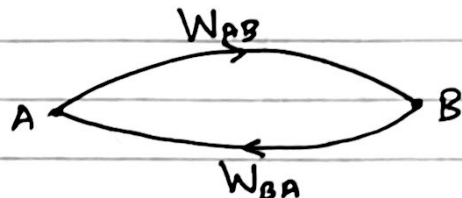
E.g.: Gravitational force, electric force, magnetic force etc.

* Non-Conservative forces

→ A force is said to be non-conservative if total work done by the force on the body over a closed path is not zero.

$$\text{i.e. } \sum W \neq 0$$

$$\text{or, } W_{AB} + W_{BA} \neq 0$$



Work done by non-conservative force depends on the length of path followed by the body.

Eg: Frictional force, Viscous force etc.

- Collision

→ The mutual interaction of bodies due to which their linear momentum and kinetic energy undergo change is called collision.

- * Types of collision

- ① Elastic collision

- The collision in which both the linear momentum and kinetic energy of the system remains conserved is called elastic collision.

So, For elastic collision,

Total linear momentum before collision = Total linear momentum after collision.

And,

Total K.E before collision = Total K.E after collision

Collision between minute particles such as collision between gases molecules, collision between atomic or subatomic particles etc. are the examples of elastic collision.

- ② Inelastic or Non-elastic collision

- The collision in which linear momentum of the system remains conserved but kinetic energy of the system does

Sn't remain conserved is called inelastic collision.

So, For Inelastic collision

Total linear momentum = Total linear momentum
before collision after collision

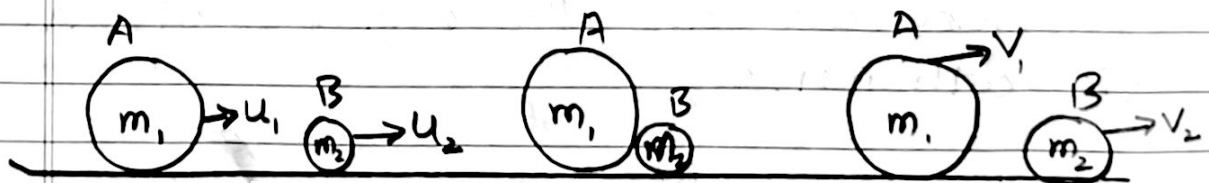
But,

Total K.E before $\neq$ Total K.E after
collision collision.

e.g. collision between larger object
Such as an collision between vehicles,
collision between rocks,
etc.

If collided bodies ~~stick~~ stick together then it is known as perfectly inelastic collision.

• One dimensional elastic collision



Consider two bodies A and B of masses m_1 and m_2 are moving in the same dir-

action with velocities u_1 and u_2 respectively. Suppose the bodies undergo elastic collision.

Let v_1 and v_2 be their velocities after collision as shown in figure.

In the elastic collision both the linear momentum and kinetic energy of the system remains conserved. So,

Total linear momentum = Total linear momentum
before collision after collision

$$\text{or, } m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$\text{or, } m_1 u_1 - m_1 v_1 = m_2 v_2 - m_2 u_2$$

$$\therefore m_1 (u_1 - v_1) = m_2 (v_2 - u_2) \rightarrow (1)$$

And.

GSS_Notes

Total K.E. = Total K.E
before collision after collision

$$\text{or, } \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$\text{or, } \frac{1}{2} m_1 u_1^2 - \frac{1}{2} m_1 v_1^2 = \frac{1}{2} m_2 v_2^2 - \frac{1}{2} m_2 u_2^2$$

$$\text{or, } \frac{1}{2} m_1 (u_1^2 - v_1^2) = \frac{1}{2} m_2 (v_2^2 - u_2^2)$$

$$\therefore m_1 (u_1 + v_1)(u_1 - v_1) = m_2 (v_2 + u_2)(v_2 - u_2) \rightarrow (2)$$

Dividing eqn (2) by eqn (1) we get,

$$\frac{m_1 (u_1 + v_1) \cancel{(u_1 - v_1)}}{\cancel{m_1 (u_1 - v_1)}} = \frac{m_2 (v_2 + u_2) \cancel{(v_2 - u_2)}}{\cancel{m_2 (v_2 - u_2)}}$$

$$\text{or, } u_1 + v_1 = v_2 + u_2$$

$$\therefore u_1 - u_2 = v_2 - v_1 \rightarrow (3)$$

This shows that; the relative velocity of approach of body before collision is equal to the relative velocity of their separation after collision.

From eqn (3),

$$v_2 = u_1 - u_2 + v_1$$

putting this value in eqn (i), we get

$$m_1(u_1 - v_1) = m_2(u_1 - u_2 + v_1 - u_2)$$

$$\text{or, } m_1(u_1 - v_1) = m_2(u_1 - 2u_2 + v_1)$$

$$\text{or, } m_1 u_1 - m_1 v_1 = m_2 u_1 - 2m_2 u_2 + m_2 v_1$$

$$\text{or, } m_1 u_1 - m_2 u_1 + 2m_2 u_2 = m_2 v_1 + m_1 v_1$$

$$\text{or, } (m_1 - m_2) u_1 + 2m_2 u_2 = (m_2 + m_1) v_1$$

$$\text{or, } v_1 = \frac{(m_1 - m_2) u_1 + 2m_2 u_2}{m_1 + m_2}$$

$$\therefore v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1 + \left(\frac{2m_2}{m_1 + m_2} \right) u_2 \rightarrow (4)$$

This gives velocity of body A after collision.

Similarly,

$$v_2 = \left(\frac{m_2 - m_1}{m_2 + m_1} \right) u_2 + \left(\frac{2m_1}{m_2 + m_1} \right) u_1 \rightarrow (5)$$

* Note: Exchange (1) & (2) in eqn (4) to get eqn (5)

* Special Case:

- If $m_1 = m_2$ then from eqn (4) & (5), we get

$$\boxed{V_1 = u_2} \text{ and } \boxed{V_2 = u_1}$$

This shows that if two bodies of equal masses undergo elastic collision then they exchange their ~~coll~~ velocities after collision.

Q.8. A ball A of mass 0.1 kg moving with a velocity of 6 m/s collides directly with a ball B of mass 0.2 kg at rest. ~~for~~

(i) Calculate their common velocity if both balls move off together.

(ii) if ~~the~~ ball A had rebounded with a velocity of 2 m/s in the opposite direction after collision, what would be the new velocity of B?

Soln

$$\begin{aligned} m_1 &= 0.1 \text{ kg} \\ u_1 &= 6 \text{ m/s} \\ m_2 &= 0.2 \text{ kg} \\ u_2 &= 0 \end{aligned}$$

Now,

$$m_1 u_1 + m_2 u_2 = m_1 v + m_2 v$$

$$\text{or, } 0.1 \times 6 + 0.2 \times 0 = 0.1 \times v + 0.2 \times v$$

$$\text{or, } 0.6 = 0.1v + 0.2v$$

$$\text{or, } 0.6 = 0.3v$$

$$\text{or, } v = \frac{0.6}{0.3} = 2 \text{ m/s}$$

ii) $V_1 = -2 \text{ m/s}$, $V_2 = ?$

Now,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$\text{or, } 0.1 \times 6 + 0.2 \times 0 = 0.1 \times (-2) + 0.2 v_2$$

$$\text{or, } 0.6 = -0.2 + 0.2 v_2$$

$$\text{or, } 0.8 = 0.2 v_2$$

$$\text{or, } v_2 = \frac{0.8}{0.2}$$

$$\text{or, } v_2 = 4 \text{ m/s}$$

$$\therefore v_2 = 4 \text{ m/s} \quad \downarrow$$

GSS_Notes

Homework / Discussion Questions

• Short Questions.

1) A man carrying a bucket of water is walking on a level road with a uniform velocity. Does he do any work on the bucket while carrying it?

→ Since displacement of man and applied force are at right angle. i.e. 90° , The work done become zero.

$$W = FS \cos \theta$$

$$= FS \cos 90$$

$$= 0 \quad \therefore \cos 90 = 0$$

GSS_Notes

$\therefore$ He doesn't do any work on the bucket while carrying it.

2) A centripetal force is acting continuously on a body moving uniformly along a circular path. Is there work done by the force. Explain.

→ Since force (gravitational force) and motion of earth are perpendicular to each other, $\theta = 90^\circ$ and so,

$$W = FS \cos \theta$$

$$= FS \cos 90$$

$$= 0 \quad \therefore \cos 90 = 0$$

Hence, there is no work done by the force.

3. "The earth moving round the sun in an orbit is acted upon by a force, hence the work must done on the earth by this force". Do you agree with this statement?

→ Earth is revolving around the sun in a circular path. For it, a centripetal force is required which is provided by gravitational force between the earth and the sun. We know, work done = $F \times s \cos \theta$, where F is gravitational force, s is distance travelled by earth and θ is the angle between them.

Since force and motion of earth are perpendicular to each other, $\theta = 90^\circ$ and work done = $F \times s \cos 90^\circ = 0$. Hence, there is no work done on the earth by this gravitational force.

4) A light body and a heavy body have the same momentum. Which one will have greater kinetic energy? Explain.

→ We know that,

$$K.E = \frac{p^2}{2m}$$

$$K.E \propto \frac{1}{m}$$

Since, kinetic energy and mass of an object are inversely proportional with each other body having less mass will have

~~high~~ greater kinetic energy.

Hence, A light body will have greater kinetic energy.

5. If the linear momentum of body increases by 50%, what is the percentage increase in its kinetic energy.

→ Soln,

let $P = p$

$$p' = p + \frac{50}{100} \times p = 1.5p$$

~~Change~~ $\frac{\%}{\%}$

Now, Kinetic energy When $p = p$

$$K.E = \frac{p^2}{2m} \rightarrow (i)$$

When $p' = 1.5p$

$$K.E = \frac{p'^2}{2m}$$

$$= \frac{(1.5p)^2}{2m}$$

$$= \frac{2.25p^2}{2m}$$

Now,

$$\text{percentage increase} = \frac{\frac{2.25p^2}{2m} - \frac{p^2}{2m}}{\frac{p^2}{2m}} \times 100\%$$

$$= 1.25 \times 100\%$$

$$= 125\%$$

∴ Increase in percentage in its kinetic energy is 125%.

6) Write the difference between elastic and inelastic collision.

→ Elastic Collision Inelastic collision

- ① The collision in which both the linear momentum and kinetic energy of the system remains conserved is called Elastic collision.
- The collision in which linear momentum of the system remain conserved but kinetic energy of the system does not remain conserved is called inelastic collision.

② A crate of mass 5 kg is lifted up from rest with uniform acceleration of 2 m/s^2 by means of vertical rope attached to it. If crate moves a vertical distance of 10 m, calculate:

- i) Work done by tension in rope to lift the crate.
- ii) Work done against gravity to lift the crate.
- iii) Net work done on crate.

→ Soln,

$$\text{mass (m)} = 5 \text{ kg}$$

$$a = 2 \text{ m/s}^2$$

$$\text{distance (s)} = 10 \text{ m}$$

i) $W = ?$

We know that,

$$\begin{aligned} W &= m(g+a) \cos \theta \times s \\ &= 5(10+2) \times \cos 0 \times 10 \\ &= 600 \text{ J} \\ \therefore W &= 600 \text{ J} \end{aligned}$$

$$\begin{aligned} \textcircled{\text{ii}} \quad W &= F s \cos \theta \\ &= m g s \cos \theta \\ &= 5 \times 10 \times 10 \times \cos 180^\circ \\ &= -500 \text{ J} \end{aligned}$$

$$\begin{aligned} \text{iii) Net work done} &= 600 \text{ J} + (-500 \text{ J}) \\ &= 100 \text{ J} \end{aligned}$$

GSS_Notes

3) You throw a 20N rock vertically into the air from ground level. You observe that when it is 15m above the ground, it is travelling at 25 m/s upward. Use the work-energy theorem to find

- i) Its speed as it left the ground
- ii) Max height.

→ Soln,

$$W_t = mg = 20 \text{ N} = F$$

$$s = 15 \text{ m}$$

$$v = 25 \text{ m/s}$$

$$\textcircled{\text{i}} \quad W = \frac{1}{2} m v^2 - \frac{1}{2} m u^2$$

$$F \cos \theta = \frac{1}{2} m v^2 - \frac{1}{2} m u^2$$

$$\text{or, } 20 \times 15 \times \cos 180 = \frac{1}{2} \times 2 \times 25^2 - \frac{1}{2} \times 2 \times u^2$$

$$\text{or, } -300 = 625 - u^2$$

$$\text{or, } u^2 = 925$$

$$\text{or, } u = \sqrt{925}$$

$$\therefore u = 30.41 \text{ m/s}$$

$$\textcircled{\text{ii}} W = \frac{1}{2} m v^2 - \frac{1}{2} m u^2$$

$$\text{or, } F \cos \theta = \frac{1}{2} m v^2 - \frac{1}{2} m u^2$$

$$\text{or, } 20 \times n \times \cos 180 = 0 - \frac{1}{2} \times 2 \times 925$$

$$\text{or, } -20n = -925$$

GSS_Notes

$$\text{or, } n = \frac{925}{20}$$

$$\therefore n = 46.25 \text{ m/s}$$